Draft version June 24, 2022
Typeset using LaTeX **twocolumn** style in AASTeX63

Occurance rate of galaxy pairs across cosmic time: Local Group analogs in a cosmological context

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ABSTRACT

Dwarf galaxies are extremely common at all redshifts and often exhibit morphological disturbances that do not primarily result from galaxy mergers. As such, pair fractions, rather than morphological indicators (asymmetry, concentration) are needed to infer dwarf galaxy merger rates from observational surveys. However, in contrast to massive galaxies ($5 \times 10^9 \leq \mathrm{M_{prim,*}} \leq 1 \times 10^{11} \mathrm{M_\odot}$), pair fractions for dwarf galaxies ($10^8 < \mathrm{M_*} < 5 \times 10^9 \mathrm{M_\odot}$) have been theoretically constrained only at low redshift. We aim to quantify the occurence rates and kinematics of isolated dwarf and massive galaxy pairs from $z = 0 - 4$ in the Illustris and IllustrisTNG cosmological simulations. Dwarf and massive subhalo pairs are identified in each snapshot of the Illustris `SUBFIND` catalogs. We find that roughly 8-10% of dwarf galaxies will have major companion (stellar mass ratio $> 1:4$) at $z = 0$, increasing steadily before plateauing in *Illustris-Dark*(*Illustris-Hydro*) at 16%(12%) by $z \sim 2$. The occurence rate of dwarf galaxies with minor companions (stellar mass ratios between $1:10$ and $1:4$) is roughly constant at $\sim 2.5\%$ from $z = 1 - 4$. In contrast, the occurence rate of major and minor massive galaxy pairs remains approximately constant across time, as massive galaxies have a major (minor) companion $\sim 18\%$ ($\sim 14\%$) of the time. This suggests that the hierarchical evolution of dwarf galaxies may proceed differently than that of massive galaxies. LMC-SMC mass analogs (dwarf minor pairs) are twice as common at redshifts greater than 2, but the present day LMC-SMC system is dynamically rare at all redshifts (in terms of pair separation and velocity). The M31-M33 pair is dynamically typical of massive minor pairs at $z \approx 0$, while the MW-LMC has unusually high velocities and low relative separations for a major massive pair at low redshift. Baryonic physics impacts the occurence rate of galaxy pairs and systematically changes the kinematics of dwarf pairs, suggesting observations of dwarf galaxy pairs can be used to test subgrid physics in cosmological simulations.

1. INTRODUCTION

Galaxy mergers are responsible for many physical processes that shape galactic evolution, including triggered star formation (Mihos & Hernquist 1996; Hopkins et al. 2013; Patton et al. 2020; Hani et al. 2020), increased AGN activity (Hopkins et al. 2008; Comerford et al. 2015; Ellison et al. 2019), and galaxy morphology (e.g., Rodriguez-Gomez et al. 2017). Dwarf galaxies make up a majority of the galaxy population at all redshifts (REF) and are also expected to undergo interactions and merge. Such processes may be important to understand the growth and evolution of intermediate mass black holes, which are often found to be "wandering" in low mass galaxy hosts (Stemo et al. 2020; Reines et al. 2020). Despite the ubiquity of dwarf galaxies in the universe, we do not have a clear path to constrain the merger rates of these systems observationally or to place known interacting dwarf galaxy systems (stellar mass $1 \times 10^8 < M_* < 5 \times 10^9 \mathrm{M_\odot}$) in a cosmological context.

Massive galaxy ($M_* > 5 \times 10^9 \mathrm{M_\odot}$) mergers and interactions can be observationally identified using methods like asymmetry and concentration measures (Conselice et al. 2000; Lotz et al. 2004). Numerous studies have used these measures to study the merger and interaction rates of massive galaxies as a function of time (e.g., Lotz et al. 2011; Casteels et al. 2014).

However, Martin et al. (2021) showed that mergers only account for 10% of morphological disturbances in dwarfs, and that a majority of their morphological disturbances derive from non-merger interactions, such as gas inflows/outflows, close passages, and fly-bys. This

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means that the typical methods of classifying interactions and mergers for massive galaxies are unlikely to be robust in lower mass regimes. This is particularly true at high redshift, when gas accretion along filaments is expected to be more significant (Kereš et al. 2009). Thus, it is critical to establish an alternative, cosmologically consistent, way to determine merger and interaction rates for dwarf galaxy pairs from observations.

Another observational technique to determine merger/interaction rates relies upon the pair fraction: the ratio of the number of host galaxies with a companion to the number of potential hosts. Pair fractions for massive galaxies have been measured observationally as a function of time (e.g., Lotz et al. 2008; Conselice et al. 2009), and calibrated against theoretical simulations to determine observability timescales that enable the translation of pair fractions to merger rates (Lotz et al. 2011; Kaviraj et al. 2015; Snyder et al. 2017; Qu et al. 2017; O'Leary et al. 2021). In contrast, pair fractions for dwarf primaries are only starting to be measured at high redshift (e.g. Ventou et al. 2017) and have only been studied with small sample size or calibrated against simulations at low redshift (Besla et al. 2018; Casteels et al. 2014; Stierwalt et al. 2015; Pearson et al. 2018). Thus we are currently missing a theoretical framework to interpret future high redshift observations of dwarf pairs, i.e. with *JWST*. In this study, we quantify the dwarf galaxy pair fraction as a function of time out to $z \sim 4$, and compare these results to the massive galaxy pair fraction, using the N-body and hydrodynamic *Illustris-1* cosmological simulations (Vogelsberger et al. 2014; Nelson et al. 2015).

There are examples of galaxy pairs involving dwarf galaxies in our Local Group, including the M31-M33, MW-LMC, and the LMC-SMC (Magellanic Clouds; MCs) systems. The frequency of such pairs have been studied extensively observationally and theoretically at low redshift. In SDSS DR7, it is found that 11% of MW/M31-like objects have one massive satellite (LMC/M33 analog) within 150 kpc and 300 km s^{-1} (Liu et al. 2011). Within 250 kpc, the frequency increases to 42% (Tollerud 2011). This is in agreement with ΛCDM predictions (Busha et al. 2011; Boylan-Kolchin et al. 2011; Patel et al. 2017; González et al. 2013). Furthermore, 4-6% of LMC/M33 analogs in both SDSS and cosmological simulations are expected to have a companion at least as massive as an SMC in the field (Besla et al. 2018; Sales et al. 2011).

However, we do not how common these pair configurations are as a function of time in the ΛCDM framework, leaving open the question as to whether our Local Group is unique in time. Recent advances in astrometry have furthermore enabled the measurement of 6D phase space information for these Local Group galaxy pairs. In this study, we will examine whether not only the frequency, but also the dynamics of pairs in our Local Group are unique *across cosmic time*.

The Local Group also contains a massive major galaxy pair (stellar mass ratio >1:4), the MW-M31. For low redshift ($z < \sim .1$) massive galaxies ($M_* > 10^{10}$ $M_\odot$), the fraction of major pairs are expected to increase as a function of mass (Casteels et al. 2014), resulting in major merger rates that rise with higher mass (Rodriguez-Gomez et al. 2015; Ventou et al. 2017; O'Leary et al. 2021). Additionally, the fraction of major pairs is also measured to increase with increasing redshift (de Ravel et al. 2009; Bundy et al. 2009; Duncan et al. 2019). In particular Duncan et al. (2019) find no strong evolution in the relative number of minor to major mergers for massive hosts out to $z<3$. But it is unclear whether these trends for the frequency of major pairs will be the same if the host mass is in the dwarf galaxy regime.

It is important to study the dwarf and massive host regimes separately, as baryonic physics, including gravitational tides, star formation, and outflows, impacts the growth of dwarf galaxies more strongly (Bullock & Boylan-Kolchin 2017; Jeon et al. 2019, 2021). For example, it has been shown that baryonic physics implemented in the *Illustris-Hydro* simulation suppresses the abundance of low mass subhalos ($\leq 10^{10} M_\odot$), but such a suppression is not present for massive halos (Chua et al. 2017). Thus, the pair fractions and dynamics of dwarf major and minor pairs may change drastically between the dark-matter-only and full-physics simulations, with less differences for massive pairs. In this study, we illustrate that the impact of baryonic physics on the growth of low mass halos in cosmological simulations may be directly testable with pair fraction observations.

Here we present the first cosmologically motivated constraints on the dwarf pair fraction for major and minor pairs as a function of time. We compare these results to those found for massive major and minor pairs, and address the differences in expected pair fractions between the N-body and full hydrodynamic simulations. This work will furthermore, self-consistently, place known galaxy pairs in our Local Group in a cosmological and temporal context. In Sec. 2, we introduce the simulations and selection criteria used to perform the analysis of our sample of pairs. In Sec. 3, we present the results of this pair fraction study, including preliminary results on the kinematics of dwarf pairs over cosmic time. In Sec. 4, we discuss the implications of our results on interpretations of future surveys, and what this could mean for constraints of feedback mechanisms in cosmo-

logical simulations. In Sec. 5 we present a summary of the main conclusions of this work, including expected pair fractions for dwarf and massive galaxy pairs.

2. SAMPLE SELECTION

Our sample of dwarf galaxy pairs and massive galaxy pairs from cosmological simulations was chosen following a strict set of selection criteria. In Sec. 2.1 we will detail the simulations from which we collected our data, Sec. 2.2 explains the abundance matching prescription used to associate dark matter halos with stellar masses, and Sec. 2.3 outlines the isolation and mass criteria used to identify our sample.

2.1. *Simulation data*

We utilize data from the *Illustris Project* (Vogelsberger et al. 2014; Nelson et al. 2015), a set of N-body and hydrodynamic cosmological simulations, to statistically analyse the occurrence rates of dwarf and massive dark matter subhalo pairs over cosmic time. In particular, we are using the highest resolution dark matter only simulation, *Illustris-1-Dark*, and the full hydrodynamic simulation, *Illustris-1* (hereafter *Illustris-Dark* and *Illustris-Hydro* respectively, see Table 1). Each of these simulations follows the dynamical evolution of 1820^3 dark matter particles from $z = 127$ to $z = 0$ through a periodic volume of 106.5^3 Mpc^3. The Illustris simulations use a cosmology that is consistent with *WMAP-9* (Hinshaw et al. 2013): $\Omega_m = 0.2726$, $\Omega_\Lambda = 0.7274$, $\Omega_b = 0.0456$, $\sigma_8 = 0.809$, $n_s = 0.963$, and $h = 0.704$.

We are utilizing data from both *Illustris-Dark* and *Illustris-Hydro* to study the effect of baryonic content on the subhalos. As dwarf subhalos have more shallow dark matter potentials than their more massive counterparts, they are more likely to be subject to disruption caused by baryonic feedback at higher redshifts, which may result in fewer low mass subhalo pairs in *Illustris-Hydro*.

We use catalogs of dark matter halos and their associated subhalos identified in each simulation using the `SUBFIND` algorithm (Springel et al. 2001; Dolag et al. 2009). Dark matter halos were identified using the Friends-of-Friends (FoF) algorithm (Davis et al. 1985), which links particles that have separations less than $0.2d$, where d is the mean inter-particle separation. Each FoF group, which we will further refer to as a "group", was further divided into over-dense and gravitationally bound dark matter subhalos.

In order to trace the growth and histories of our subhalos, we are using the merger trees provided by the `SUBLINK` algorithm (Gómez et al. 2015), which tracks subhalos between snapshots using their particle data. These merger trees provide information about the kinematics and mass growth of subhalos in both *Illustris-Dark* and *Illustris-Hydro*, and in particular will be used to provide the maximal masses of each of the subhalos we investigate.

The lower mass threshold for subhalos that constitute our dwarf pairs have a halo mass greater than $1\times10^9 \mathrm{M}_\odot$, such that each of our halos are resolved into at least 130 particles in *Illustris-Dark* and 150 particles in *Illustris-Hydro*.

	Illustris-Dark	Illustris-Hydro
$\mathrm{L_{Box}}$ [Mpc]	106.5	106.5
$\mathrm{m_{DM}}$ [$\mathrm{M_\odot}$]	7.5×10^6	6.3×10^6

Table 1. Parameters of the Illustris simulations used in this analysis.

2.2. *Abundance matching*

Although *Illustris-Hydro* provides stellar masses for all subhalos, however these reported masses are higher than observations suggest (Genel et al. 2014). We assign stellar masses to dark matter halos in a consistent way between both *Illustris-Dark* and *Illustris-Hydro*using the abundance matching (AM) prescription of Moster et al. (2013) to associate each dark matter halo with a stellar mass.

The AM relationship at low stellar masses has a large spread, thus a subhalo with a dark matter halo mass of 10^{11} may be associated with a stellar mass anywhere between $1.2\times10^8 - 5\times10^9 \mathrm{M}_\odot$, as can be seen in Fig. **??**.

There are two inputs to the AM relationship: dark matter mass and redshift. As shown recently by Munshi et al. (2021), the relationship between stellar mass and peak halo mass of dwarf subhalos in the Marvel and Justice League simulations is much tighter than the relationship between the stellar mass and halo mass of subhalos at $z = 0$. We use the maximal dark matter mass of the halo to ensure that the stellar mass associated with each subhalo reflects the shape of the dark matter profile in the inner regions of the subhalo, thus preventing us from assigning very low stellar masses to halos that have been stripped of their outer dark matter mass by recent interactions. We evaluate the AM relation using the redshift of the halo at the snapshot that we are interested in, rather than the redshift of the halo at its maximal mass, in order to account for ongoing star formation past peak halo mass.

For the collection of subhalos identified at each snapshot of the simulation, the stellar masses are drawn from

	Dwarf Pairs	Massive Pairs
Group Mass	$8 \times 10^{10} < \mathrm{M_g} < 5 \times 10^{11} \mathrm{M}_\odot$	$1 \times 10^{12} < \mathrm{M_g} < 4 \times 10^{12} \mathrm{M}_\odot$
Subhalo Current Mass	$> 1 \times 10^{10} \mathrm{M}_\odot$	$> 5 \times 10^{11} \mathrm{M}_\odot$
Primary	$1 \times 10^{8} < \mathrm{M}_* < 5 \times 10^{9} \mathrm{M}_\odot$	$5 \times 10^{9} < \mathrm{M}_* < 1 \times 10^{11} \mathrm{M}_\odot$

Table 2. Selection criteria for dwarf and massive pairs.

the gaussian sampled AM relation. We thus generate 50 realizations for the stellar mass distribution of the same subhalo population in order to account for uncertainties in the AM relation. Each realization is treated as an independent sample of galaxies, where computed properties (stellar masses, relative speeds, etc) are averaged over each realization. We report the mean of all realizations and error bars indicate the 1σ dispersion across the realizations.

2.3. *Selection criteria*

In this section, we will outline the selection criteria used to identify dwarf subhalo and massive subhalo pairs in the *Illustris-Dark* and *Illustris-Hydro* simulations. In short, we first make a series of mass cuts on the group mass to constrain the masses of the halos that may be contained in the group and to ensure that the systems are relatively isolated. Then, the subhalos of each identified isolated group undergo further mass cuts until we are left with a sample of both dwarf and massive subhalo pairs. A summary of the mass criteria used for our selection can be found in Table 2.

2.3.1. *Isolation*

We are interested in the occurrence rates of subhalo pairs in pristine environments to ensure that the dynamics of the subhalos in groups are unaffected by nearby massive perturbers.

FoF groups are large collections of nearby associated particles throughout the Illustris simulation. As there is no boundedness criteria for these groups, we wanted to ensure that pulling pairs of subhalos from FoF groups would yield a sample of subhalos that were dynamically unperturbed by more massive nearby structures. We calculated the isolation fraction

$$f_{\mathrm{iso}} = N_{\mathrm{Hill}}/N_{\mathrm{FoF}} \tag{1}$$

where N_{Hill} is the number of pairs of subhalos that satisfy the Hill radius criterion (see Appendix 5), and N_{FoF} is the number of pairs from FoF groups, and found that $> 99.8\%$ of all pairs identified in FoF groups are considered isolated by the more stringent Hill radius criterion. We thus used these FoF groups as a proxy for isolation, and considered *all* pairs from FoF groups isolated.

2.3.2. *Group mass cuts*

We first make a series of cuts on the group masses reported in the *Illustris-Dark*and *Illustris-Hydro*catalogs at each snapshot out to $z \sim 12$. Since we are interested in subhalo pairs in two different mass regimes, we will use different mass criteria to isolate groups that may contain dwarf pairs, and groups that may contain more massive galaxy pairs. As mentioned, we do not search for dwarf pairs in massive groups since the central subhalos of these more massive groups would perturb the dwarf system.

Group mass cuts are performed on the group mass provided in the SUBFIND catalogs, `Group_M_TopHat200`, which is the total mass enclosed in a sphere whose mean density is $\rho_c \Delta_c$. The **dwarf groups** must have a group mass $8 \times 10^{10} < \mathrm{M_g} < 5 \times 10^{11} \mathrm{M}_\odot$ while **massive groups** have mass $1 \times 10^{12} < \mathrm{M_g} < 4 \times 10^{12} \mathrm{M}_\odot$. These dwarf and massive group masses are roughly representative of Magellanic Cloud analogs (Besla et al. 2012; Moster et al. 2013) and Local Group analogs respectively (van der Marel et al. 2012a).

In addition to group mass cuts, we ensure that each group has at least one subhalo that satisfies the following properties:

- *dwarf* - maximal halo mass is $8 \times 10^{10} < \mathrm{M_{max}} < 5 \times 10^{11} \mathrm{M}_\odot$ with mass at the snapshot of interest $\mathrm{M_{snap}} > 1 \times 10^{10} \mathrm{M}_\odot$

- *massive* - maximal halo mass is $1 \times 10^{12} < \mathrm{M_{max}} < 4 \times 10^{12} \mathrm{M}_\odot$ with mass at the snapshot of interest $\mathrm{M_{snap}} > 5 \times 10^{11} \mathrm{M}_\odot$.

This ensures that each group has at least one subhalo that will serve as a primary galaxy for the catalog after stellar mass cuts are applied (Sec. 2.3.3).

We only consider subhalos in each of these groups that have subhalo masses $M_H > 10^9 M_\odot$ at the given snapshot. This ensures that each dwarf pair will consist of subhalos that are resolved into more than 100 dark matter particles at the snapshot of interest.

In practice we find that some low mass groups consist of a set of subhalos with only one or two subhalos with mass above our lower mass threshold of $10^9 \mathrm{M}_\odot$, while the remaining subhalos have lower mass. We refer to

these types of groups as "One Subhalo" and "Two Subhalo" groups respectively, since they contain either one or two candidate primaries. Groups with three or more subhalos passing this lower mass threshold are called "Three+ Subhalo" groups.

2.3.3. *Dwarf and Massive Primaries*

Dwarf and massive primaries are defined to be the most massive subhalo in each group with the following properties:

- *dwarf* - stellar mass between $10^8 \leq \mathrm{M_{prim,*}} \leq 5 \times 10^9 \mathrm{M_\odot}$
- *massive* - stellar mass between $5 \times 10^9 \mathrm{M_{prim,*}} \leq 1 \times 10^{11} \mathrm{M_\odot}$.

We make mass cuts using the stellar masses of the subhalos in order to best compare our results with observational studies.

The average number of dwarf and massive primary subhalos is plotted as a function of redshift in Fig. **??**.A subset of this data can be found in Table 3. Also plotted is the number of primaries in groups with the marked number of companions.

We find that the number of dwarf primaries increases from $z = 4$ to about $z \sim 0.5$, at which point the number of primaries decreases by approximately 15% approaching $z = 0$. There are significant differences in the number of dwarf primaries that belong to One- and Two-Subhalo Groups in *Illustris-Dark* and *Illustris-Hydro*. *Illustris-Hydro* boasts about 2× more primaries in One-Subhalo groups, 1.5-2× more primaries in Two-Subhalo Groups, and 0.6× fewer Three-Subhalo groups than *Illustris-Dark*.

The number of massive primaries increases consistently from high to low redshift. There are very few massive groups with only one or two large ($\mathrm{M_H} > 10^9 \odot$) subhalos, since groups of this mass scale typically host at least one large subhalo and its companions and satellites. Thus, almost 100% of the massive primaries reside in "Three-Subhalo" groups.

2.3.4. *Pairs*

The different pair types are defined to contain a dwarf or massive primary subhalo with a secondary subhalo:

- *Major Pairs*: with stellar mass $M_{\mathrm{sec},*}$ such that $1/4 \leq M_{\mathrm{sec},*}/M_{\mathrm{prim},*} \leq 1$.
- *Minor pairs*: with stellar mass $M_{\mathrm{sec},*}$ such that $1/10 \leq M_{\mathrm{sec},*}/M_{\mathrm{prim},*} \leq 1/4$.
- *Minor+ Pairs*: with stellar mass $M_{\mathrm{sec},*}$ such that $M_{\mathrm{sec},*}/M_{\mathrm{prim},*} \leq 1/4$. This set of pairs *includes* the whole set of minor pairs, but also includes pairs whose mass ratio is less than 1/10.

The MCs, as well as the MW+LMC system and M31+M33 system, have a stellar mass ratio of approximately 1/10, and thus would be classified as minor pairs according to our analysis. All pairs must be separated by at least 15 physical kpc so that the pairs are well resolved into individual component subhalos in `SUBFIND`.

3. RESULTS

After collecting our sample of dwarf and massive pairs, we analyzed the frequency of occurrence as well as their properties, such as mass ratio and average kinematics of the pairs, as a function of time. By comparing results for groups identified in *Illustris-Dark* and *Illustris-Hydro*, we can compare the effect that feedback prescriptions may have on the formation of low mass dwarf pairs vs. massive pairs, and search for differences in their kinematics, which ultimately dictate their merger rates.

Although we collected pairs out to redshift $z \sim 12$, the number of identified primaries drops significantly approaching $z \sim 4$. As such, the following analysis will focus only on a redshift range of $z = 0- \sim 4$, where the statistics are more meaningful. In Sec. 3.1, we will discuss the stellar, current halo, and maximum halo mass ratios for our sample, which will enable us to examine the mass evolution of these systems. In Sec. 3.2, we present the main data product of this paper, namely the pair fractions for dwarf and massive major and minor pairs in *Illustris-Dark* and *Illustris-Hydro*. In Sec. 3.3, we discuss our findings on the kinematics of our population of pairs.

3.1. *Stellar and Halo Masses and Ratios*

In Fig. **??**, we show the average stellar masses of the primaries of major and minor dwarf and massive pairs identified in *Illustris-Dark* and *Illustris-Hydro*. We find that the average stellar masses of primaries increases over time, typically peaking at or near $z = 0$. The average stellar mass of the primaries of minor and minor+ *dwarf* pairs is much greater than for major pairs across both simulations. There are two explanations for this effect. Firstly, if many major pairs merge by low redshift, leaving behind a more massive central halo and smaller minor satellites, the expected stellar mass of primaries of minor pairs would be significantly higher than the stellar masses of primaries in major pairs. Secondly, because of the stellar mass floor applied to the dwarf sample, only the largest dwarf primaries are contained in the set of minor and minor+ pairs. However, the average stellar mass of major and minor massive pairs is roughly similar. Since the lower mass floor for massive secondaries

	Dwarfs				Massive			
z =	0	1	2	4	0	1	2	4
Isolated Groups	#	#	#	#	#	#	#	#
Primaries								
Pairs								
Major Pairs								
Minor Pairs								
Minor+ Pairs								

Table 3. Placeholder for now Number of groups, primaries, and pairs

is lower than for primaries (see Sec. 2.3), minor massive pairs may have lower mass primaries than possible for the same range of stellar mass ratios for dwarfs.

The stellar and dark matter mass ratios between galaxies or subhalos impact the timescale for the interactions that will occur between pairs owing to dynamical friction (in addition to kinematics, see Sec. 3.3). Interacting subhalos of roughly equal mass may end up tearing each other apart and recombining as a completely new composite subhalo. However, when a large subhalo interacts with a much less massive minor companion, the minor companion may disrupt, forming stellar streams around the more massive primary, with very little disruption to the primary. Separating our collection of pairs by mass ratios will allow us to explore dynamics and interactions separately in these two regimes (Cox & Loeb 2008).

In the following discussion, stellar mass ratios will refer to the AM stellar mass of the secondary companion subhalo over the AM stellar mass of the primary. Halo mass ratios refer to the subhalo mass of the secondary reported in the simulation catalog at that redshift, over the subhalo mass of the primary at that redshift. Maximum (max) halo mass ratio refers to the maximum or "peak" subhalo mass ever achieved by the secondary subhalo, over the peak subhalo mass achieved by the primary at any point in their evolution. Thus, stellar and halo mass ratios are calculated using values of the stellar and halo masses at the current snapshot, while the max mass ratio may use subhalo masses for the primary and secondary from different snapshots.

We present the median stellar mass ratios, current halo mass ratios, and maximum mass ratios for dwarf and massive major and minor pairs in Fig. **??**. Since pairs are selected based on their stellar mass ratios in each of the stellar mass realizations, the median stellar mass ratios of major and minor pairs is relatively flat across all redshifts and shows little difference between *Illustris-Dark* and *Illustris-Hydro*.

Since the stellar mass of each subhalo is assigned based on the subhalo's maximal mass, we find, as expected, that the max mass ratio to evolve with redshift in the same way as the AM relationship for a fixed stellar mass ratio.

However, the halo mass ratio may tell us about how these halos have evolved since they achieved their peak halo masses, including owing to interactions. In particular, we find that in all cases the halo mass ratio falls well below the max halo mass ratio, implying that the secondaries may be losing much of their peak halo mass, potentially to the primaries, thus reducing the halo mass ratio at each redshift.

The time evolution of the The halo mass ratio for dwarf pairs increases as a function of time, which could indicate that a number of high mass ratio (minor) pairs have merged by low redshift. However, this is not the case for massive galaxies. Rather, the median halo mass ratio remains approximately constant throughout the considered redshift range.

To understand the evolution of max mass ratios for massive pairs, it is important to remember that the abundance matching relationship is quite shallow at high subhalo masses and low redshifts. This means that, for a fixed stellar mass ratio, the possible corresponding halo ratio drops, as shown by the max halo mass ratio of massive pairs dipping below the stellar mass ratio at low redshift in Fig. **??**.

3.2. *Pair fractions*

Pair fractions provide insight as to how likely it is to find two galaxies very nearby each other in observations. Pair fractions can be used to back out merger rates from observations where other traditional methods of determining merger rates, such as asymmetry and concentration parameters, may fail. However, as pair fractions have only been calibrated as a function of time for massive pairs, it is important to separately determine expected pair fractions for dwarf pairs, rather than simply extrapolating the results for massive pairs.

After each group's primary was identified, we considered the second most massive halo by stellar mass, and sorted the pairs by stellar mass ratio. The pair fraction

was calculated as

$$F_{\rm pair} = \frac{N_{\rm comp}}{N_{\rm prims}},$$

where $N_{\rm comp}$ is the number of pairs with a companion of a certain type (i.e., a major or minor companion), and $N_{\rm prims}$ is the number of primaries. These values are computed for each stellar mass realization independently, and are then averaged for each snapshot. In Fig.**??**, we show the pair fractions of different pair types as a function of time out to $z \sim 4$. The shaded regions represent the 1σ error on the average of the pair fraction over time, computed by conducting 50 realizations of the stellar catalog at each redshift by sampling the AM relation.

We found that dwarf primaries are much more likely to have at least one major companion at redshifts $z > 2$ than at $z = 0$: approximately 10% (8%) of primaries in *Illustris-Dark* (*Illustris-Hydro*) have a major companion at high redshifts ($z > 2$), though this fraction drops to 4% by $z = 0$ in both simulations. The fraction of primaries with a minor companion remains approximately constant from redshift $z = 0 - 4$: ∼3-4%(∼5%) of primaries have a minor (minor+) companion in both *Illustris-Dark* and *Illustris-Hydro*.

There is a slight suppression (between 1-4%) in the dwarf pair fractions in *Illustris-Hydro*. As can be seen in Fig. **??**, *Illustris-Hydro* contains many more One-subhalo groups than *Illustris-Dark*, the number of potential primary subhalos with no companions is much higher, resulting in a decrease in the pair fractions across all pair types between *Illustris-Dark* and *Illustris-Hydro*.

The pair fraction of massive subhalos evolves dramatically differently than their dwarf counterparts, with the pair fraction either remaining constant or *increasing* at low redshifts, and is much higher than the fraction of dwarf primaries with companions. In particular, the fraction of massive primaries with at least one major companion in *Illustris-Dark* remains approximately constant for redshifts of $z = 0-4$ at between $15-20\%$, while the same pair fraction in *Illustris-Hydro* increases from $\sim 12\%$ to 20% by $z = 0$.

There is a slight difference between the behavior of the fraction of massive primaries in *Illustris-Dark* and *Illustris-Hydro* for "All Pairs" and major pairs. In particular, there is a sharp uptick in the fraction of primaries with these types of companions at $z \sim 1$ in only *Illustris-Dark*. This fraction then declines to z ∼0.2, then rises sharply to $z = 0$. While the sharp rise at low redshift is seen in *Illustris-Hydro* as well, the sharp increase at $z = 1$ is not present. This means that there is a significant difference in the fraction of primaries expected with a major companion between the two simulations. In particular, the fraction of major pairs peaks at $z = 0$ in *Illustris-Hydro*, while the fraction of major pairs in *Illustris-Dark*is smaller than at $z = 1 - 3$. This simulation-dependent behavior is not present in the minor pairs.

There is a difference in the relative pair fractions of major and minor dwarf pairs at high redshift, since there is such a dramatic suppression of the fraction of primaries with a major companion. If this can be observationally constrained, it may help us to test the baryonic physics implemented in cosmological hydrodynamic simulations.

3.3. *Kinematics*

We analyzed the average separations and relative velocities of dwarf and massive identified pairs in *Illustris-Dark* and *Illustris-Hydro*. We found that there are discrepancies between the kinematics of pairs in the two simulations.

In all cases of massive and dwarf pairs, *Illustris-Hydro* boasts typically much higher relative velocities ($\sim 20 -$ 40km/s higher) between major pairs than *Illustris-Dark*. While this velocity separation is less extreme for minor pairs ($\sim 5-15$km/s higher), it is still evident, especially at higher redshifts ($z > 2$).

3.3.1. *Separations*

Because this study utilizes subhalos selected in cosmological simulations, we can extract the full 6D-phase information for subhalos, such as 3D position and velocity vectors. We thus do not have to worry about projection effects artificially introducing projected pairs to our dataset. This also means that we can study the evolution of the physical separations of the subhalo pairs and how it might be impacted by mass scale or simulation physics.

In Fig. **??**, we show the average separation of our selection of pairs (major and minor pairs, dwarf and massive galaxy pairs) selected in both the *Illustris-Dark* and *Illustris-Hydro*simulations.

We find that the average separation of all pair types increases towards lower redshift. This could indicate that the small separation pairs are merging, leaving behind the population of pairs that have longer-lived orbits. Alternatively, the population of pairs picked out with our selection criteria may be very different between lower and higher redshift. Following high redshift pairs and establishing their orbits through to $z = 0$ will be the subject of future work.

There is a slight (∼20 kpc), but systematic, separation difference between major pairs in *Illustris-Dark* and *Illustris-Hydro* in both of the dwarf and massive galaxy regimes. When we considered only subhalos that had

separations > 80 kpc, the separation difference was resolved between the two simulations, indicating that a low separation population of pairs is absent in *Illustris-Hydro*. This population of subhalos may have merged earlier in the simulation. This will be studied in future orbital studies.

On average, minor pairs have much lower average separations than major pairs. This trend holds true between both simulations and both mass ranges. Interestingly, the separation discrepancy does not appear for minor pairs, which suffer less dynamical friction, supporting the hypothesis that major close pairs are being missed or merging faster in the *Illustris-Hydro*simulation.

3.3.2. *Velocities*

Figure **??** illustrates the magnitude of the mean relative 3D velocities between all pair types in both simulations. The relative velocities decrease significantly throughout time for all pairs, where massive pairs have speeds of order 100 km/s larger than dwarf pairs.

There is a significant ∼15km/s (∼30-50km/s) velocity offset between major dwarf (massive) pairs in *Illustris-Dark* versus *Illustris-Hydro*. Additionally, a small (∼5-10km/s) velocity separation exists between minor pairs. While this velocity separation is suppressed at very low redshifts for both pair types, the vast spread between velocities suggests a significant orbital/dynamic evolution between pairs in the simulations.

When we exclude all pairs with relative separations < 80kpc from our analysis, the stark velocity differences between *Illustris-Dark* and *Illustris-Hydro* are resolved, implying that the "missing" population of low separation halos in *Illustris-Hydro* is also responsible for this velocity offset.

In the future, we will include studies of the tangential vs. radial velocities to provide a first order guess about the orbital trajectories of the these systems to determine if they on average are on infalling or radial orbits.

4. DISCUSSION

We analyzed a set of dwarf galaxy pairs and massive galaxy pairs from the *Illustris-1* simulation to place constraints on the cosmologically expected fractions of major and minor pairs as a function of time. Our goals include identifying any differences in the behavior of pairs as a function of primary mass (dwarf or massive galaxy) and simulation physics. Here we use our results to place Local Group Pairs in a cosmological and temporal context (Sec. 4.1). We further discuss implications of our findings for cosmological simulations with baryonic physics (Sec. 4.2). Finally, we briefly compare to existing literature (Sec. 4.3).

4.1. *Implications for Local Group Pairs*

Patel et al. (2017) showed that the existence of the MW-LMC and M31-M33 pairs, are typical of minor massive pairs at low redshift. However, while the M31-M33 interaction may be characteristic of minor massive pairs, they show that the MW and LMC interaction is dynamically unique, since very high velocity and low separation pairs are dynamically rare at $z = 0$. We found that typical minor massive pairs have separations between 200-300kpc and relative velocities between 160-200kpc at low redshifts (z <0.5). Since the M31-M33 pair has a current separation of ∼200kpc and relative velocity ∼200km/s, and the MW-LMC pair has a separation of ∼50kpc and relative velocity of ∼320km/s, our results are in good agreement with Patel et al. (2017). However, while the MW-LMC kinematics may not be typical at low redshift, they are more characteristic of higher redshift ($z > 2$) minor massive pairs, where separations are on average <150kpc and velocities exceed 240km/s.

Since the MCs are at a separation of only $23kpc$ and have a relative velocity of $103km/s \pm 26$ Zivick et al. (2018), the combination of separations and velocities is non-typical, as this separation is uncommon at low redshift, while these velocities are uncommon at higher redshift. These kinematic differences can be potentially reconciled by the presence of the MW, and the study of the dynamics of triples will be the topic of future work.

While Besla et al. (2018) finds that 70-85% of dwarf pairs in their sample are major pairs, we find that between 40-45% of all pairs are major pairs, although this fraction increases at higher redshift to ∼60-65%. The stellar mass range of our study includes stellar masses down to $10^8 M_\odot$ while the Besla et al. (2018) study includes stellar masses down to 2×10^8. Our lower stellar mass limits and ability to ensure that pairs are associated in the same group may increase the number of dwarf minor pairs in our sample.

4.2. *Impact of Mass Scales and Baryonic Physics*

The mass scale of our galaxy pairs heavily influences the behaviour of the pair fractions as a function of time. The pair fractions of dwarf major pairs are ∼4% at z=0 in both simulations, however the pair fraction grows steadily and peaks at $z = 2$, where the values plateau. This time-dependent evolution is very different from our results for massive pairs, since massive major pair fractions stay roughly constant or *decrease* at higher redshift. This indicates that the evolution of dwarf and massive galaxy pairs proceeds very differently and on different timescales.

We also found that the evolution of dwarf pair fractions between $z = 0 - 4$ is dramatically different be-

	Dwarf		Massive		Local Group Analogs			
	Major Pairs	Minor Pairs	Major Pairs	Minor Pairs	MW-M31	M31-M33	MW-LMC	LMC-SMC
$M_{*,\mathrm{prim}}$	#	#	#	#	#	#	#	#

Table 4. Placeholder for now Summary table with all values from z=0 (and maybe z=1,2,3,etc)

tween *Illustris-Dark* and *Illustris-Hydro*. In particular, the pair fractions in *Illustris-Hydro* are approximately $20-25\%$ lower than those of *Illustris-Dark*. This suppression effect is not seen with massive pairs, implying that the implementation of baryonic physics in *Illustris-Hydro* impacts the evolution of dwarf pairs much more strongly than massive pairs.

4.3. *Comparison to existing work*

Lotz et al. (2008) finds that the merger fraction of massive major galaxy pairs ($M_* > 10^{10} M_\odot$) is approximately constant at $10\pm2\%$ between $0.2 < z < 1.2$, which is consistent with our findings that massive major pair fraction evolves very little (only $\pm2\%$) as a function of redshift. While the redshift evolution of our pairs is consistent with Lotz et al. (2008), we do find higher pair fractions across this same redshift range (with pair fractions ranging between 16-20%), which may be due to our increased sample of massive major pairs at high separation.

Casteels et al. (2014) found that the major merger fraction for galaxies with $10^{9.5} < M_* < 10^{11.5} M_\odot$ (roughly equivalent to our massive pair criteria) is approximately 1.3-2% at low redshift, but increased to 4% at lower masses. We find that massive major pairs have a pair fraction of $f_{pair} \sim 18-20\%$ at low redshift, which is inconsistent with the Casteels et al. (2014) work, but close to 4% for dwarf galaxy major pairs, which is consistent with observations. However, Casteels et al. (2014) uses morphological signatures of mergers to calculate merger fractions. Since massive galaxies will only have morphological signatures from recent close passages of other massive galaxies, these methods will preferentially select close pairs compared to larger separation pairs. Since our study includes many low redshift, highly separated pairs, it is not unreasonable to assume that the major pair fraction difference between the two studies arises from these selection criteria.

Duncan et al. (2019) found that the observational massive major pair fraction increases with redshift from $z = 0$ (where $f_{pair} \sim 0.07$) to $z = 4$ ($f_{pair} \sim 0.2$). This is in contrast to our finding that the massive major pair fraction is constant/slightly decreasing from $z = 0-4$ (from $f_{pair} \sim 0.2$ to $f_{pair} \sim 0.14$)). This is likely due to the strict projected separation cuts that are made on the observational data, where massive major pairs must have $5 < r_{proj} < 30kpc$, while our massive major pairs includes low separation *and* high separation (r_{sep} >30kpc) pairs. This accounts for our inflated pair fraction at all redshifts compared to the Duncan et al. (2019) work.

In general, our study finds inflated pair fraction results because of our ability to determine associated, but well separated, dwarf and massive galaxy pairs.

5. CONCLUSIONS

In this paper we presented a study of isolated pairs of dwarf and massive subhalos in the large volume cosmological simulation *Illustris-1*, using data from both the N-body (dark matter only) and full hydrodynamic simulations, *Illustris-Dark* and *Illustris-Hydro* respectively. Our criteria requires that all pairs are a part of the same FoF group (see Sec. 2.3). We ensure isolation of our selection of pairs using a Hill Radius criteria to ensure that pairs are not gravitationally influenced by nearby massive perturbers.

We define dwarf galaxies as having stellar mass $10^8 M_\odot < M_* < 5\times10^9 M_\odot$, and massive galaxies as $5\times10^9 < M_* < 10^{11} M_\odot$. Major pairs have secondary to primary stellar mass ratios between 1-1:4, minor pairs have stellar mass ratios between 1:4-1:10, and minor+ pairs have stellar mass ratios <1:4, thus including the population of minor pairs.

We found that there is a difference in the evolution of the pair fraction for dwarf pairs and massive pairs over time, and that baryonic physics has an impact on the expected pair fractions and kinematics of these systems.

Our pair fraction results are summarized as follows:

(i.) **The pair fractions for all dwarf primaries with a companion with stellar mass $> 10^8 M_\odot$ (All Pairs) peaks at $z = 2$ and remains constant to $z = 4$.** In *Illustris-Dark*, the pair fraction is approximately 8% at $z = 0$ and increases to peak at 16% at $z = 2$, while in *Illustris-Hydro*, the pair fraction is 8% at $z = 0$, increasing to ∼12% at $z = 2$. In both simulations, the trends with redshift are determined by the major pairs, which are the most common class of dwarf galaxy pair at all redshifts.

(ii.) **Magellanic Cloud mass analogs (dwarf minor pairs) are approximately 1.5x more common at $z > 2$ than at the present day.** In *Illustris-Hydro*, ∼2-3% of dwarf primaries (LMC analogs) host a minor

companion (SMC analog) across all redshifts, with only a slight increase in the pair fraction between $z = 0$ and $z = 2$. There is slightly more evolution in the case of *Illustris-Dark*, with ∼4% of primaries hosting a minor companion for $z > 2$, and declining to 2% at $z = 0$. Thus, we expect to find isolated minor dwarf pairs with approximately the same frequency out to $z = 4$.

(iii.) **Dwarf primaries are 2-2.5× more likely to have major companions at $z > 2$ than at $z = 0$.** We expect that roughly ∼4% of dwarf primaries will have a major companion at $z = 0$, increasing to $z = 2$ before leveling off at 10% (8%) in *Illustris-Dark* (*Illustris-Hydro*) for $z > 2$.

(iv.) **The pair fraction of dwarf primaries with companions of all types (major, minor, minor+) is between $1-4\%$ less common in *Illustris-Hydro* than in *Illustris-Dark*.** This indicates that the baryonic physics of the *Illustris-Hydro*simulation has a non-negligible impact on the expected pair fractions for low mass systems. The same suppression of pair fractions for these pair types is not seen for massive pairs.

(v.) **The pair fractions for all massive galaxy primaries with a companion with a stellar mass $> 10^8 M_\odot$ increases over time and peaks between $z = 0-1$.** This behaviour is primarily driven by the pair fraction of massive minor+ pairs, as the pair fractions for massive major and minor pairs is relatively flat across $z = 0-4$. This is very different than the behaviour of low mass galaxy pairs, where the major dwarf pairs drive the evolution of pair fractions for dwarf pairs.

(vi.) **∼15% of MW/M31 analogs host a LMC/M33 mass analog at all redshifts (minor pairs for massive galaxies).** The behavior is similar in *Illustris-Dark*and *Illustris-Hydro*. It is also similar to the behavior of dwarf galaxy minor pairs (LMC/SMC analogs) with time.

(vii.) **The frequency of MW-M31-type pairs (major massive galaxy pairs) may be impacted by baryonic physics,** as demonstrated by differences in the time evolution of pair fractions between *Illustris-Dark* and *Illustris-Hydro*. In the case of *Illustris-Dark*, the pair fraction of massive major pairs is ∼17-18% at $z = 0$, and increases to roughly 20% between $z = 1-3$. However, in *Illustris-Hydro*, the pair fraction of massive major pairs peaks at 20% at $z = 0$, and declines steadily to 15% at $z = 4$. The evolution of major galaxy pairs is markedly different than for dwarf galaxies

(viii.) **These results are/are not consistent with observations.**

Our results for pairs in terms of their kinematics are as follows:

(i.) **The relative separation between dwarf (massive) galaxy major pairs increases by ∼100kpc (∼250kpc) from $z = 4$ to $z = 0$.** The minor pairs show similar trends with the relative separations increasing by ∼90kpc (∼180kpc) from $z = 4$ to $z = 0$. There is a relative separation difference between major pairs in *Illustris-Dark* vs. *Illustris-Hydro*of approximately 10-15kpc (20-40kpc), with average separations higher across all redshifts in *Illustris-Hydro*. At $z = 0$, the average separation of dwarf minor pairs is ∼140-150 in both simulations.

(ii.) **The relative velocities of between dwarf (massive) galaxy major pairs decreases by ∼30km/s (∼90km/s) from $z = 4$ to $z = 0$.** The dwarf (massive) minor pairs show similar trends, where the average velocity decreases by 40km/s (100km/s) from $z = 4$ to $z = 0$. There is a significant impact on the average velocities of pairs in this sample from baryonic physics, as there is a velocity separation between the two simulations of approximately 20km/s (40km/s) for dwarf (massive) major pairs, and 10km/s (20km/s) for dwarf (massive) minor pairs.

(iii.) **The MCs are dynamically unique compared to the population of dwarf minor pairs at $z = 0$.** Their separation is much smaller and their velocities much larger than is typical for dwarf minor pairs.

(iv.) **While the M31-M33 pair is dynamically typical of massive minor pairs at low redshift, the MW-LMC pair is extremely unique both in terms of its small separation and large velocities.**

(v.) **MW-M31 pair has lower relative velocities and higher separations than typical massive major pairs at low redshift.**

We've shown that there is a difference in pair fractions between dwarf pairs and massive pairs. In addition, baryonic effects implemented in *Illustris-Hydro* have a non-negligible impact on the behaviour of the evolution of pair fractions and on the typical kinematics, and we'll be able to see how these discrepancies play out in the era of *JWST*.

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APPENDIX

Hill Radius Criteria The Hill radius is the distance from the center of a group, G, at which circular orbits of particles within G will no longer be stable due to the presence of a massive perturber P.

The Hill radius is defined as:

$$R_{\rm Hill} = R_{\rm sep}\left(\frac{M_{\rm G}}{3M_{\rm P}}\right)^{1/3}, \tag{1}$$

where $M_{\rm P}$ is the mass of the perturber, $M_{\rm G}$ is the mass of the group under consideration, and $R_{\rm sep}$ is the physical separation between the two groups (Hahn et al. 2009). łAs the Hill radius is a measure of the distance from center that particles in G can sustain circular orbits, we consider the group G isolated if the radius of the group is less than the Hill radius. The radius of the group is taken to be `Group_R_TopHat200` provided in the `SUBFIND` catalog, which is the radius of a sphere, centered on the group, whose mean density is $\rho_c\Delta_c$, where ρ_c is the critical density of the universe and Δ_c is given by the spherical top-hat collapse model of Bryan & Norman (1998). This ensures that the particles belonging to G are gravitationally bound to G and that the intrinsic dynamics of the subhalo pairs contained within are not affected by massive perturbers.